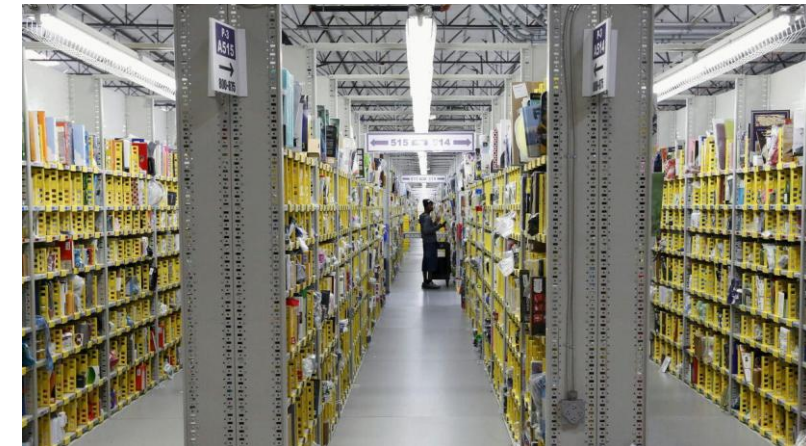


# 1. Introduction

## 1.2 Data-Generating Processes

### What can be forecast?

Forecasting is required in many situations: deciding whether to build another power generation plant in the next five years requires forecasts of future demand; scheduling staff in a call center next week requires forecasts of call volumes; stocking an inventory requires forecasts of stock requirements.

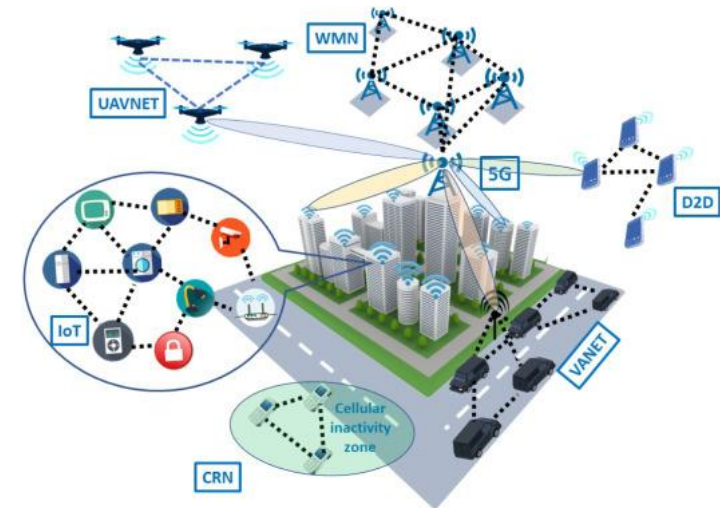
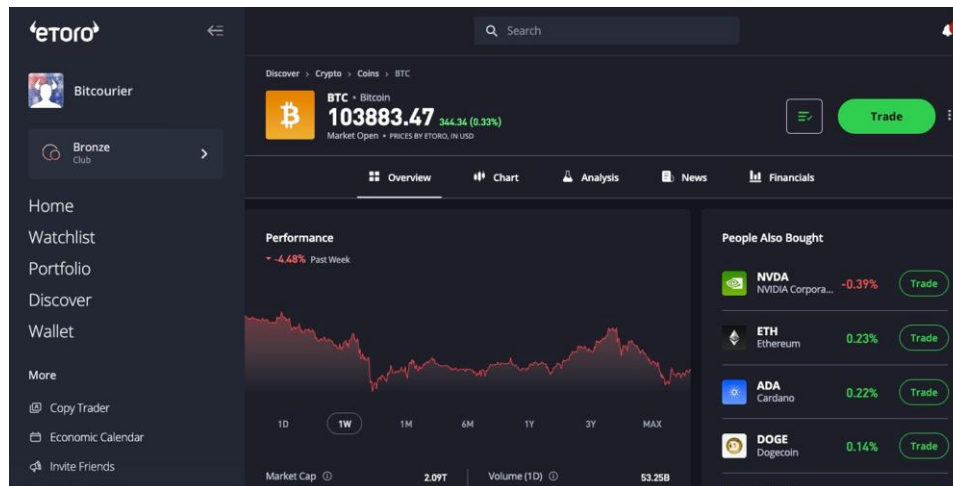


# 1. Introduction

## 1.2 Data-Generating Processes

### What can be forecast?

Forecasts can be required several years in advance (for the case of capital investments), or only a few minutes beforehand (for telecommunication routing). Whatever the circumstances or time horizons involved, forecasting is an important aid to effective and efficient planning.



# 1. Introduction

## 1.2 Data-Generating Processes

### What can be forecast?

Some things are easier to forecast than others. The time of the sunrise tomorrow morning can be forecast precisely. On the other hand, tomorrow's lotto numbers cannot be forecast with any accuracy. The predictability of an event or a quantity depends on several factors including:

1. how well we understand the factors that contribute to it;
2. how much data is available;
3. how similar the future is to the past;
4. whether the forecasts can affect the thing we are trying to forecast.



# 1. Introduction

## 1.2 Data-Generating Processes

### **What can be forecast?**

For example, short-term forecasts of residential electricity demand can be highly accurate because all four conditions are usually satisfied.

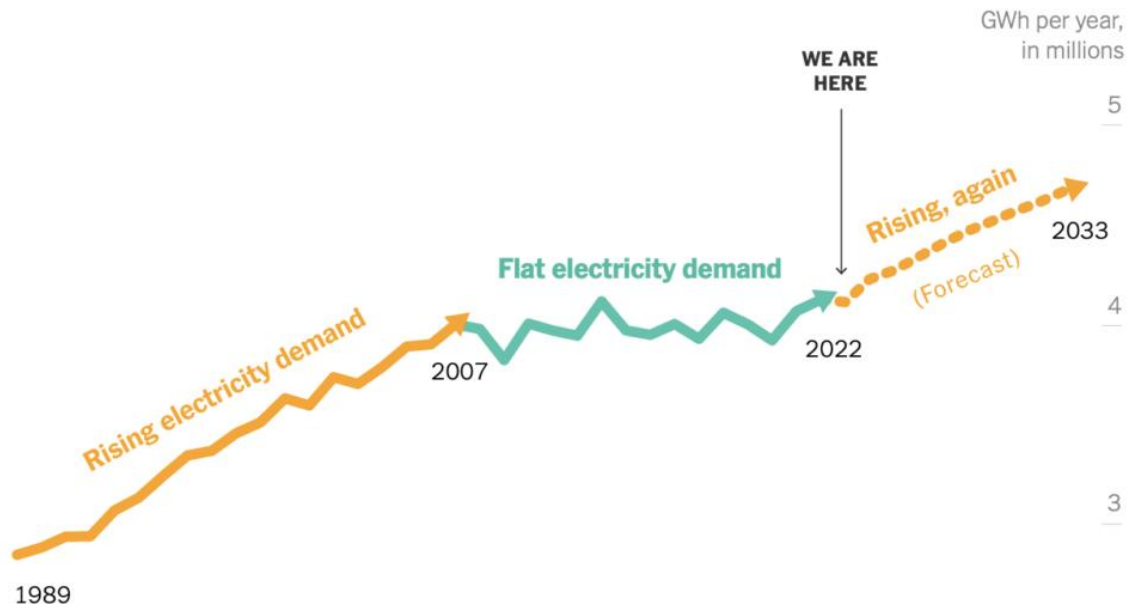
1. We have a good idea of the contributing factors: electricity demand is driven largely by temperatures, with smaller effects for calendar variation such as holidays, and economic conditions.
2. Several years of data on electricity demand are usually available, and many decades of data on weather conditions.
3. For short-term forecasting (up to a few weeks), it is safe to assume that demand behavior will be similar to what has been seen in the past.
4. For most residential users, the price of electricity is not dependent on demand, and so the demand forecasts have little or no effect on consumer behavior.

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## 1.2 Data-Generating Processes

### What can be forecast?

Provided we have the skills to develop a good model linking electricity demand and the key driver variables, the forecasts can be remarkably accurate. Check references.



<https://everlightsolar.com/surge-in-power-use-across-the-united-states-is-concerning/>

<https://www.nerc.com/programs/reliability-assessment--performance-analysis/electricity-supply--demand>

<https://isonewswire.com/2025/05/01/enhanced-long-term-forecast-predicts-steady-growth-in-energy-use-peak-demand/>

# 1. Introduction

## 1.2 Data-Generating Processes

### What can be forecast?

On the other hand, when forecasting currency exchange rates, only one of the conditions is satisfied: there is plenty of available data. However, we have a limited understanding of the factors that affect exchange rates, the future may well be different to the past if there is a financial or political crisis in one of the countries, and forecasts of the exchange rate have a direct effect on the rates themselves. If there are well-publicised forecasts that the exchange rate will increase, then people will immediately adjust the price they are willing to pay and so the forecasts are self-fulfilling.



| Currency                   | Bank Buys Notes | Bank Sells Notes |
|----------------------------|-----------------|------------------|
| US Dollar USA              | 31.51           | 32.8             |
| Singapore Dollar Singapore | 23.46           | 24.5             |
| 日本円 (: 100) Japan          | 25.83           | 28.08            |
| 人民币 China                  |                 |                  |

#### Exchange Rate Formula

$$\text{Exchange Rate} = \frac{\text{Money in After Exchange}}{\text{Money Before Exchange}}$$

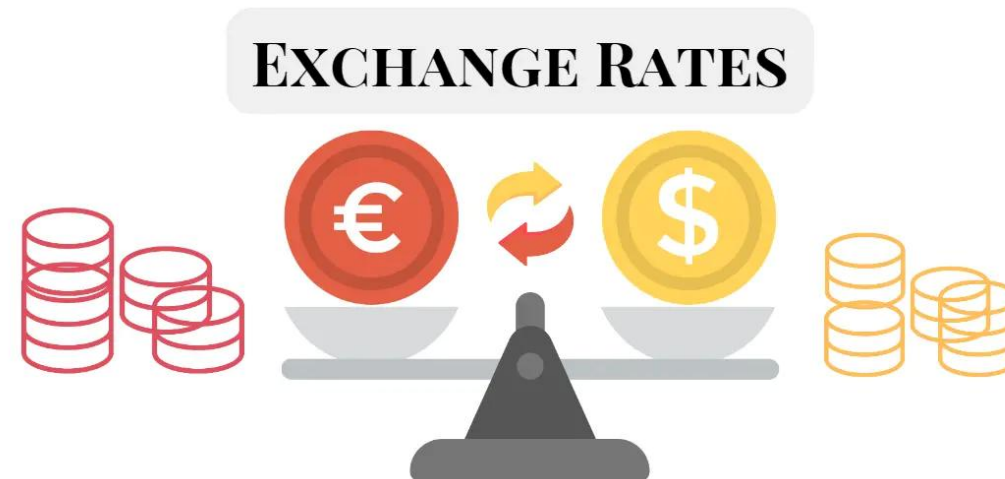


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## 1.2 Data-Generating Processes

### What can be forecast?

In a sense, the exchange rates become their own forecasts. This is an example of the “efficient market hypothesis”. Consequently, forecasting whether the exchange rate will rise or fall tomorrow is about as predictable as forecasting whether a tossed coin will come down as a head or a tail. In both situations, you will be correct about 50% of the time, whatever you forecast. In situations like this, forecasters need to be aware of their own limitations and not claim more than is possible.



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## 1.2 Data-Generating Processes

### What can be forecast?

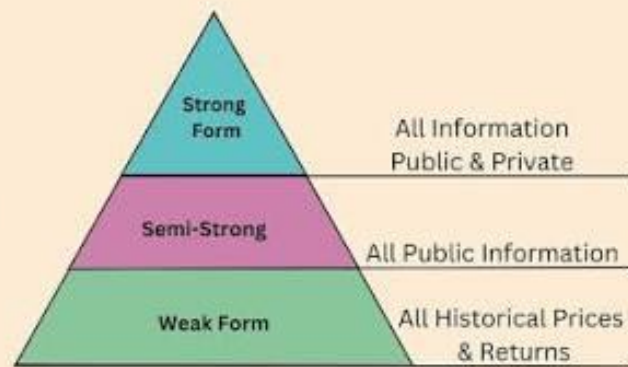
#### Efficient Market Hypothesis (EMH)

##### Can You Beat the Market?

The Efficient Market Hypothesis (EMH) suggests that **financial markets are highly efficient** and that stock prices always reflect all available information. This makes it **nearly impossible** for investors to outperform the market over the long term consistently.

But how  
does EMH  
work?

Let's break it  
down into  
three forms





# 1. Introduction

## 1.2 Data-Generating Processes

### What can be forecast?

#### MEXICO IN NUMBERS: CHANGES IN THE USD:MXN EXCHANGE RATE THIS CENTURY

##### THE NUMBERS YOU NEED TO KNOW



The peso lost nearly half its value this century.

##### Between Dec. 29, 2000 and Aug. 5, 2026:



the Mexican peso depreciated **44.3%**  
against the US dollar



or, put the other way, the greenback  
appreciated **79.6%** over that stretch.

##### The all-time extremes are 16.18 pesos apart.

**25.12**

The century's weakest peso was **25.12** to the dollar on March 24, 2020, just 13 days after the WHO declared COVID-19 a pandemic.

**8.94**

The strongest was **8.94** on May 21, 2001.

**64.4%**

From peak to trough, that's a **64.4%** peso depreciation (a **181%** dollar appreciation).

Source: Banxico

Mexico News  
DAILY

<https://mexiconewsdaily.com/business/mexico-in-numbers-peso-exchange-rate-history-2000-dollar/>

# 1. Introduction

## 1.2 Data-Generating Processes

### What can be forecast?

Often in forecasting, a key step is knowing when something can be forecast accurately, and when forecasts will be no better than tossing a coin. Good forecasts capture the genuine patterns and relationships which exist in the historical data, but do not replicate past events that will not occur again.



# 1. Introduction

## 1.2 Data-Generating Processes

### What can be forecast?

Many people wrongly assume that forecasts are not possible in a changing environment. Every environment is changing, and a good forecasting model captures the way in which things are changing. Forecasts rarely assume that the environment is unchanging. What is normally assumed is that the way in which the environment is changing will continue into the future. That is, a highly volatile environment will continue to be highly volatile; a business with fluctuating sales will continue to have fluctuating sales; and an economy that has gone through booms and busts will continue to go through booms and busts.

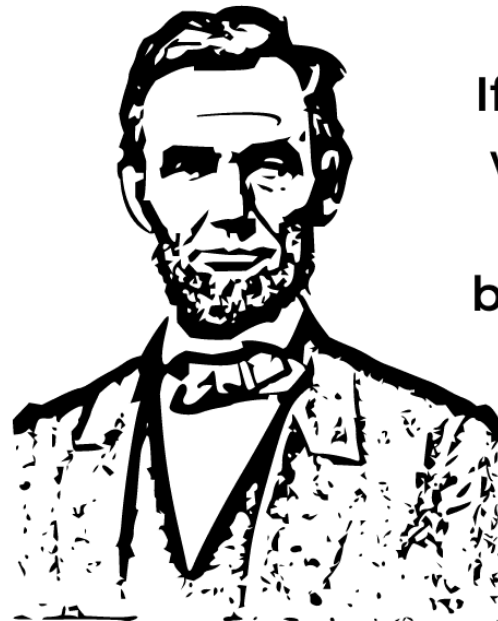


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## 1.2 Data-Generating Processes

### What can be forecast?

A forecasting model is intended to capture the way things move, not just where things are. As Abraham Lincoln said, “If we could first know where we are and whither we are tending, we could better judge what to do and how to do it”.



If we could first know where  
we are, and whither we are  
tending, we could then  
better judge what to do, and  
how to do it.

Abraham Lincoln

# 1. Introduction

## 1.2 Data-Generating Processes

### **What can be forecast?**

Forecasting situations vary widely in their time horizons, factors determining actual outcomes, types of data patterns, and many other aspects. Forecasting methods can be simple, such as using the most recent observation as a forecast, or highly complex, such as neural networks. Sometimes, there will be no data available at all. For example, we may wish to forecast the sales of a new product in its first year, but there are obviously no data to work with. The choice of method depends on what data are available and the predictability of the quantity to be forecast.

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## 1.2 Data-Generating Processes

### What can be forecast?



Forecasting is a common statistical task in business, where it helps to inform decisions about the scheduling of production, transportation and personnel, and provides a guide to long-term strategic planning

# 1. Introduction

## 1.2 Data-Generating Processes

### What can be forecast?

Forecasting is about predicting the future as accurately as possible, given all the information available, including historical data and knowledge of any future events that might impact the forecasts.

Forecasting should be an integral part of the decision-making activities of management, as it can play an important role in many areas of a company. Modern organisations require short-term, medium-term and long-term forecasts, depending on the specific application.



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## 1.2 Data-Generating Processes

### Important Remarks

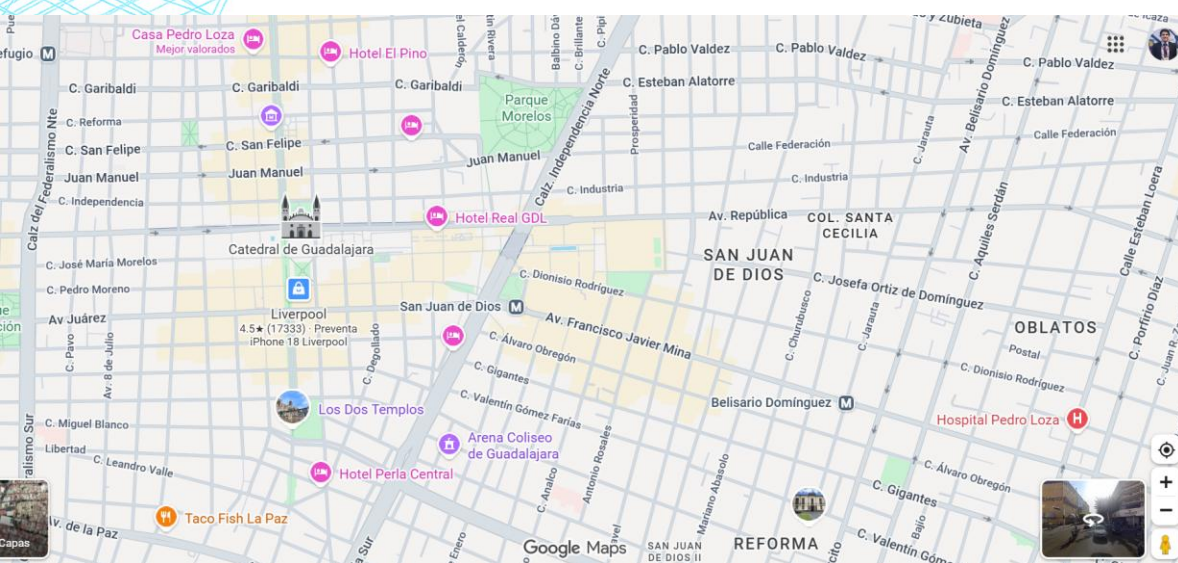
- Time series data is a collection of observations made sequentially along the time dimension.
- Any time series is generated by some kind of mechanism (ex. Shipments of a manufacturing plant, exchange rates, online market inventories, etc).
- Time series data is produced by stochastic and deterministic processes.
- The deterministic processes involve quantities that evolve in a predictable manner over time.
- A stochastic process is a way to describe how things change over time in a random but somewhat predictable manner (like how the weather changes daily with some patterns and probabilities involved)



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## 1.2 Data-Generating Processes

### Important Remarks

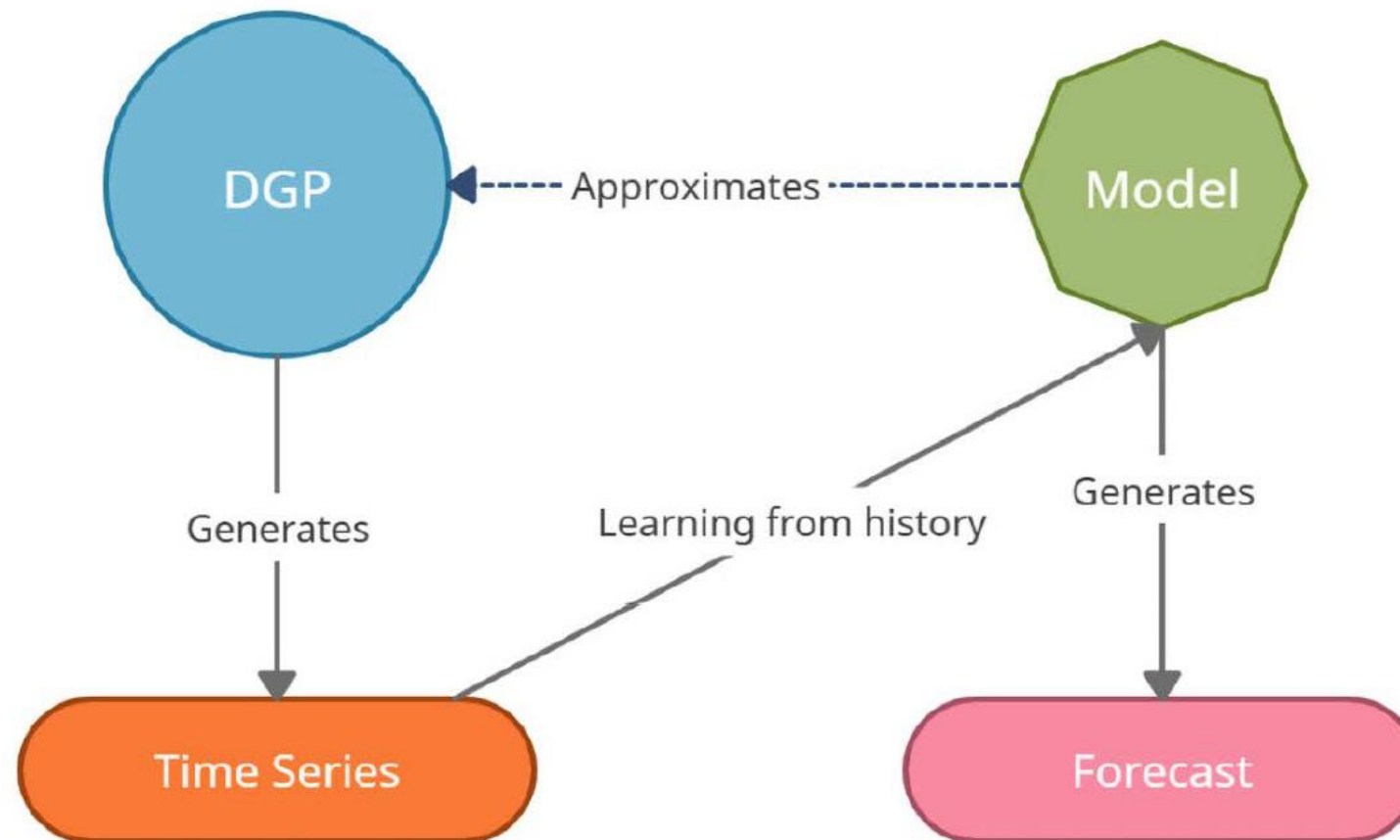


The map of GDL is certainly useful. But a map of GDL is not the same as a photo of GDL. It doesn't showcase the danger, nightlife or the complex traffic. A map is just a model that represents some useful features of a location, such as roads and places.

# 1. Introduction

## 1.2 Data-Generating Processes

### Important Remarks



# 1. Introduction

## 1.2 Data-Generating Processes

### **Important Remarks**

Students Reflection

When we consider that we have a useful model ?